

ZAOYANG, China

WMO: 572790

Lat: 32.1411N

Lon: 112.7553E

Elev: 127

StdP: 99.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.4	-14.5	1.1	3.6	-11.5	1.4	3.3	6.3	3.8	5.4	4.1	1.3	40	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	35.6	26.0	34.4	25.5	33.3	25.1	28.0	32.9	27.5	32.2	26.9	31.5	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.8	30.6	26.2	22.0	30.3	25.6	21.2	29.9	91.0	32.9	88.5	32.4	86.0	31.7	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.7	4.1	-6.1	37.5	1.5	1.2	-7.2	38.3	-8.1	39.1	-8.9	39.7	-10.0	40.6		
			-7.0	29.0	1.5	0.7	-8.1	29.5	-9.0	29.9	-9.9	30.3	-11.0	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	3.4	6.2	11.7	17.5	22.7	26.5	28.5	27.6	23.4	18.1	11.5	5.6
	DBStd	9.34	3.13	3.97	4.39	4.15	3.32	2.73	2.67	3.05	3.37	3.52	3.95	3.03
	HDD10.0	523	204	117	32	2	0	0	0	0	0	0	28	140
	HDD18.3	1733	462	340	209	62	5	0	0	0	3	48	207	396
	CDD10.0	3060	1	10	85	228	393	496	574	546	402	251	72	3
	CDD18.3	1228	0	0	4	38	139	246	316	287	155	41	1	0
	CDH23.3	12482	0	0	26	277	1159	2558	3771	3242	1214	227	7	0
	CDH26.7	5262	0	0	3	68	414	1067	1782	1456	429	43	0	0
Wind	WSAvg	1.8	1.7	1.8	2.1	2.1	2.1	2.0	2.0	1.8	1.7	1.5	1.6	1.6
Precipitation	PrecAvg	862	15	23	48	83	100	102	164	133	88	58	36	12
	PrecMax	1255	43	88	116	414	268	266	319	347	352	163	130	45
	PrecMin	548	0	0	3	7	21	9	33	30	4	0	2	0
	PrecStd	184	11	19	27	76	46	65	84	88	68	38	29	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	19.7	26.5	31.2	34.2	36.0	37.4	37.2	35.2	30.6	24.4	16.7
		MCWB	7.7	12.2	16.7	20.2	21.0	23.5	27.5	26.8	23.2	19.4	15.0	9.1
	2%	DB	12.5	16.8	23.2	28.6	32.4	34.4	35.7	35.3	32.6	27.7	21.8	14.2
		MCWB	6.6	9.9	15.0	19.2	20.6	23.5	27.0	26.6	23.1	17.8	13.7	8.0
	5%	DB	10.6	14.7	20.8	26.6	30.7	33.1	34.5	33.8	30.8	26.0	19.6	12.4
		MCWB	5.5	8.8	13.8	18.2	20.3	23.3	26.7	26.3	22.2	17.3	12.9	7.1
	10%	DB	8.7	12.6	18.7	24.5	29.0	31.7	33.3	32.5	29.1	24.1	17.5	10.7
		MCWB	4.6	7.6	12.6	17.1	20.0	23.3	26.4	25.9	21.6	16.9	12.2	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	13.4	18.5	22.6	24.6	27.6	29.0	28.7	26.7	21.6	16.7	10.6
		MCDB	13.2	17.5	24.1	27.5	29.6	32.5	33.9	33.0	32.7	25.9	20.9	13.8
	2%	WB	7.4	11.1	16.4	20.9	23.2	26.5	28.2	28.0	25.5	20.0	15.2	9.2
		MCDB	11.1	15.7	21.3	26.0	27.9	31.2	33.3	32.6	30.0	24.7	19.4	12.7
	5%	WB	6.1	9.6	14.7	19.5	22.3	25.7	27.6	27.4	24.4	19.0	14.1	8.1
		MCDB	9.8	13.0	19.2	24.6	27.2	30.1	32.6	32.0	28.3	23.4	18.1	11.7
	10%	WB	4.9	8.4	13.2	18.2	21.4	24.9	27.1	26.8	23.1	17.9	12.9	6.8
		MCDB	8.1	11.8	17.8	23.2	26.5	29.1	31.9	31.3	26.7	22.4	16.6	9.9
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	8.6	9.0	8.8	7.8	6.8	7.0	7.6	7.8	7.5	7.4
		MCDBR	10.9	11.8	11.6	12.0	11.7	9.9	8.4	8.7	9.9	10.8	10.7	10.7
		MCWBR	6.3	6.4	5.5	5.0	3.7	2.9	2.5	2.6	3.1	3.7	5.0	6.0
	5% WB	MCDBR	9.7	9.8	10.6	10.4	8.7	7.7	7.5	7.5	8.3	8.6	8.9	9.2
	MCWBR	6.1	5.9	5.6	5.3	3.9	3.0	3.0	2.8	3.7	3.8	4.6	5.6	
Clear-Sky Solar Irradiance	taub	0.642	0.702	0.703	0.696	0.687	0.733	0.702	0.734	0.722	0.709	0.680	0.631	
	taud	1.538	1.466	1.484	1.540	1.601	1.549	1.642	1.564	1.554	1.522	1.536	1.579	
	Ebn at Noon	545	566	616	651	664	632	649	618	597	553	512	523	
	Edh at Noon	232	273	286	281	266	279	254	271	264	254	230	212	
All-Sky Solar Radiation	RadAvg	2.12	2.50	3.65	4.33	4.59	4.68	4.64	4.34	3.55	2.97	2.39	2.07	
	RadStd	0.32	0.55	0.34	0.43	0.43	0.46	0.60	0.56	0.48	0.41	0.34	0.35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.32	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page